Cost of Capital
Practice Problems

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These two problems will help you practice using CAPM, calculating a firm's weighted average cost of capital, and doing NPV.

1. You've just been hired by McKinsey to evaluate a new project being considered by one of its client firms. Someone else at McKinsey has already calculated that the project's free cash flows will be -$100,000 in year 0, $70,000 in year 1, and $60,000 in year 2. Is this a positive NPV project? To answer this, you need to find the company's cost of capital, WACC. Assume that the company has 1.8 million shares of stock outstanding. The stock sells for $40 a share. The firm's debt is publicly traded and was recently quoted as 90% of face value, and its face value is $60 million. The YTM on this debt is 12%. Assume that the risk-free rate is 3.97%, and the market risk premium is 7.6%. The company has a beta of 0.60, and corporate tax rate of 34%.

    a. Using the CAPM model, what is the firm's cost of equity?

    b. What is the total market value of equity?

    c. What is the total market value of debt?

d. Using your answers from (a) – (c), what is the firm's WACC?

e. Given your answer to part (d), and assuming the project's risk is like the company's existing activities, should it undertake the project?

2. Moxi, Inc. has 4 million outstanding shares that sell at price of $15, and their beta equals 1.5. The company has a $4 million loan from Bank of America with an interest rate of 7 percent, and $20 million par value of debt that is priced 90 percent of par and has a yield to maturity of 11 percent. The risk-free rate is 5 percent, and market premium is 6 percent. The corporate tax rate is 35%.

    a. What is Moxi's market value of equity?

    b. What is Moxi's market value of debt?

    c. What is Moxi's cost of equity?

    d. What is Moxi's cost of debt?

    e. Using your answers from parts (a)-(d), what is Moxi's weighted average cost of capital (WACC)?